

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)



ck800wa

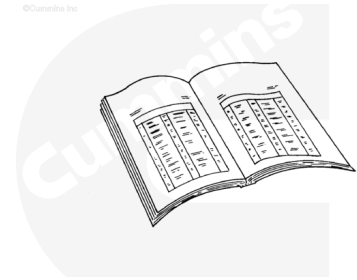
with Electronically Actuated Injector

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

- Remove the coolant bypass tube. Refer to Procedure 008-005 in Section 8
(/qs3/pubsys2/xml/en/procedures/28/28-008-005-tr.html) (right bank **only**).
- Remove the exhaust gas temperature converters, if equipped. Refer to Procedure 019-450 in Section 19 in the Troubleshooting and Repair Manual QSK38, QSK50, and QSK60 Modular Common Rail System, Bulletin 4021533.
(/qs3/pubsys2/xml/en/procedures/05/05-019-450.html)
- Remove the intake manifold air. Refer to Procedure 019-059 in Section 19 in the Troubleshooting and Repair Manual QSK38, QSK50, and QSK60 Modular Common Rail System, Bulletin 4021533.
(/qs3/pubsys2/xml/en/procedures/05/05-019-059.html)
- Remove the intake manifold air pressure sensor. Refer to Procedure 019-061 in Section 19 in the Troubleshooting and Repair Manual QSK38, QSK 50, and QSK60 Modular Common Rail System, Bulletin 4021533.
(/qs3/pubsys2/xml/en/procedures/05/05-019-061.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)

Remove

with Mechanically Actuated Injector

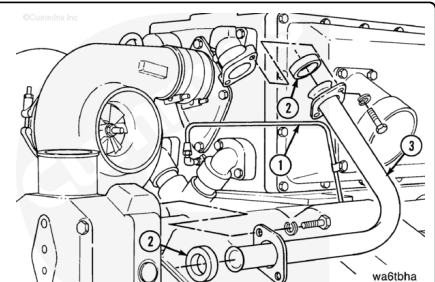
Center Mount Aftercooler

Remove the eight capscrews. Remove the two tubes (3).

Remove the rectangular ring seals (2).

Discard the seals.

Detach the air fuel control air tube (1) at the aftercooler.



wa6tbha

Note : If the engine has a hose connection between the turbocharger and aftercooler housing, remove the T-bolt clamps and slide the hose back.

Note : If the turbocharger uses turbo retaining straps, remove the turbocharger. Refer to Procedure 010-033 in Section 10.

(/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html)

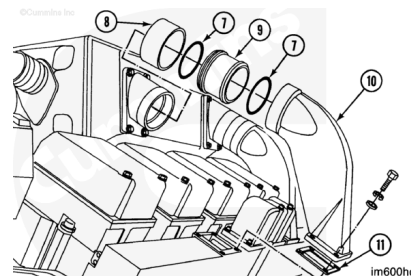
Note : There are four air crossovers.

Remove the four capscrews. Pull the crossover (10) and the tube (9) from the aftercooler.

Remove the gasket. Remove the tube from the crossover. Remove the o-rings (7).

Discard the o-rings.

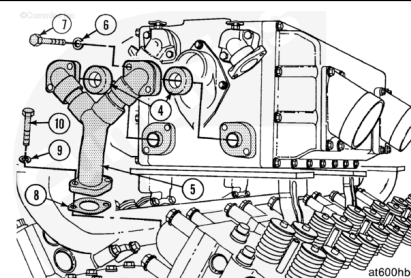
Remove the dust seal (8). If the dust seal is hard or soaked with oil, discard the seal.



Remove the two capscrews (10). Remove the four capscrews (7). Remove the water inlet connection assembly (5).

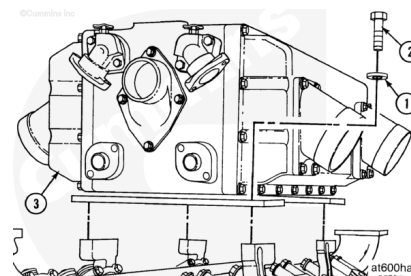
Remove the two rectangular ring seals (4). Remove the gasket (8).

Discard the gasket and seals.



⚠ WARNING ⚠

This assembly or component weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : Older engines have eight support brackets to the support leg capscrews (2)

Remove the 12 capscrews (2).

Remove the aftercooler assembly (3).

Outboard Aftercooling - Standard Cooling system

Old style plumbing with the hoses removed.

Note : Each of the four aftercoolers has a coolant inlet and outlet tube. Only the tubes for the left bank front are shown.

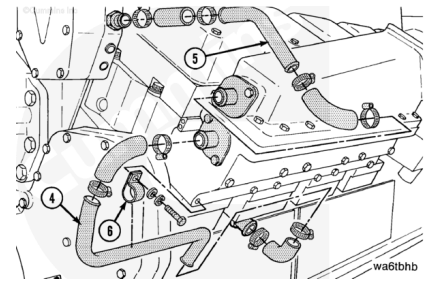
Remove both tubes from all four of the aftercoolers.

Note : The rocker lever housing end of the outlet tube for the left bank rear aftercooler has a tube nut and a rectangular seal instead of a hose and hose clamps.

Remove the capscrew for any support clamp (6).

Loosen the hose clamps.

Remove the inlet (11) and outlet (12) tubes. Remove the hoses.



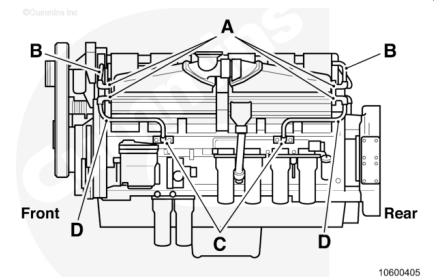
New Style Plumbing

Note : Each of the four aftercoolers has a coolant inlet and outlet tube. Only the tubes for the left bank are shown

Remove both tubes from all four of the aftercoolers.

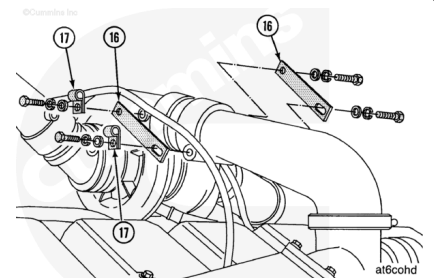
Remove the capscrews for any support clamp.

Loosen the tube nuts and seals (A). Discard the seals. Loosen the hose clamps at the cylinder block connection (C). Remove the inlet (8) and outlet (D) tubes. Remove the hoses.

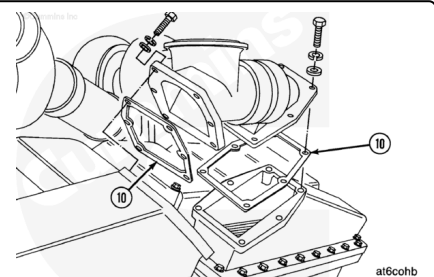


Type I - Crossovers with Turbo Retaining Straps

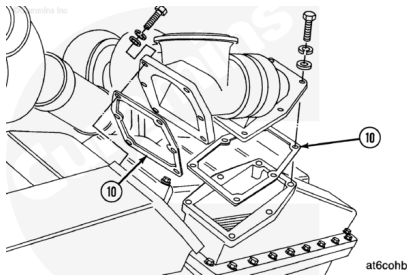
Remove the capscrews, the clamps (17) for the turbocharger oil inlet hose, and the retaining straps (16).



Remove the V-band clamp (15). Pull the air crossover (14) and tube (12) from the turbocharger. Remove the dust seal (13), the tube (12) and the o-ring seals.



Remove the capscrews, the aftercooler-to-aftercooler connection assembly and the gaskets.



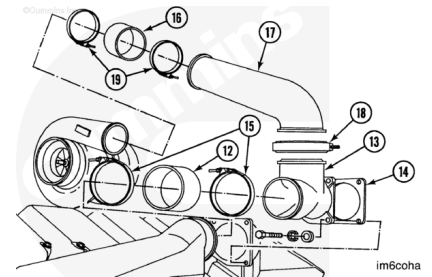
Type II - Air Crossovers with Hose Connections

Loosen all of the hose clamps.

Remove the V-band clamp (18). Pull the air crossover (17) and the hose (16) from the turbocharger. Push the hose (12) on the connection (13) as far as possible.

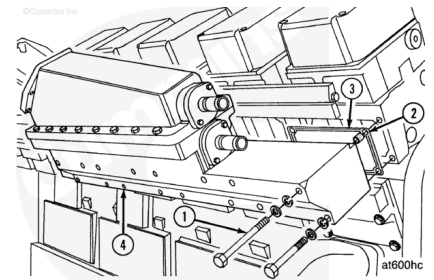
Remove the four capscrews. Remove the connection and gasket. Remove the hose from the connection.

Discard the hoses and gasket.



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Note : If the aftercooler water tubes have not been removed, remove them at this time

Remove all of the capscrews. Remove the aftercooler assembly (40) and the gaskets. Remove the rectangular seal sleeves (2).

Discard the seals and gaskets.

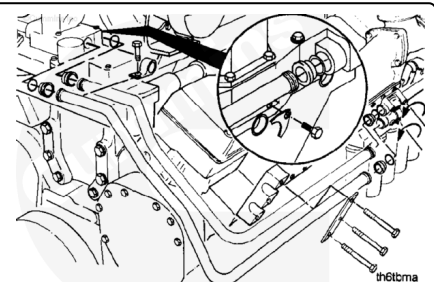
Outboard Aftercooler - Low Temperature Aftercooling (LTA) Cooling System

The following procedure should be used in both engine banks.

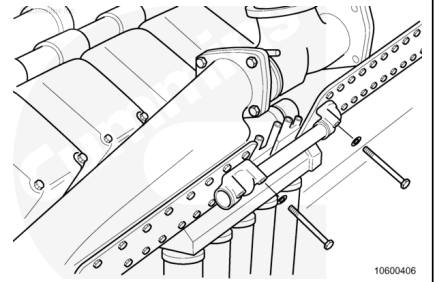
Loosen capscrews on the water transfer connections to create flexibility.

Remove the capscrews for any supporting clamp.

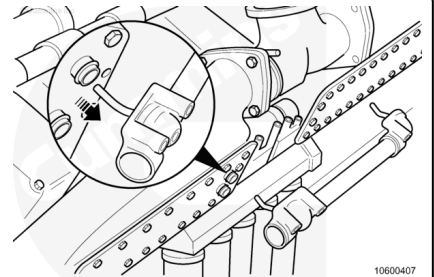
Remove the water transfer tubes.



Remove the capscrews on the water transfer connections between the aftercooler cores.



Remove the water transfer connections.



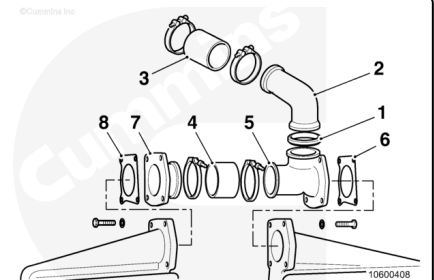
Loosen all of the hose clamps.

Remove the V-band clamp (1). Pull the air crossover (2) and the hose (3) from the turbocharger. Push the hose (4) on the connection (5) as far as possible.

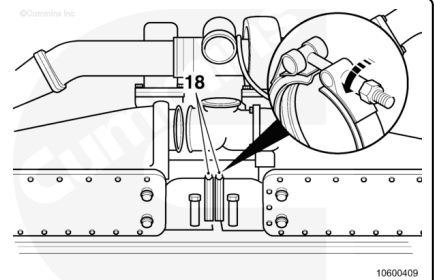
Remove the four capscrews. Remove the connection and gasket (6). Remove the hose from the connection.

Remove the four capscrews. Remove the flange (7) and the gasket (8).

Discard the hoses and gaskets.

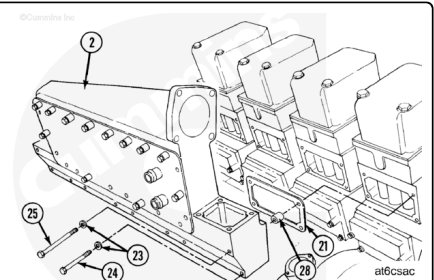


Loosen hose clamps (18).



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Remove all of the capscrews. Remove the aftercooler assembly (2) and the gaskets.

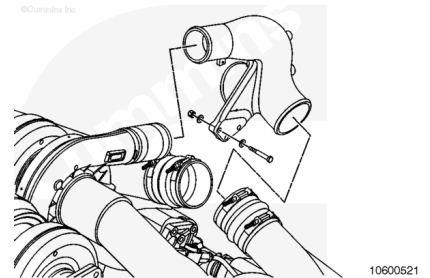
Remove the rectangular seal sleeves (28).

Discard the seals and gaskets.

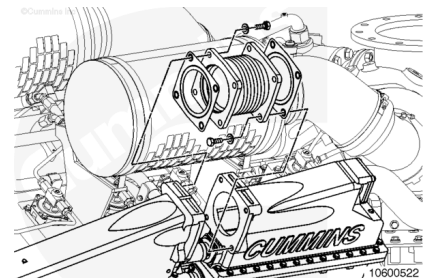
with Electronically Actuated Injector

Remove the hose clamps at the air crossover to turbocharger hose connection and at the aftercooler cover hose connections.

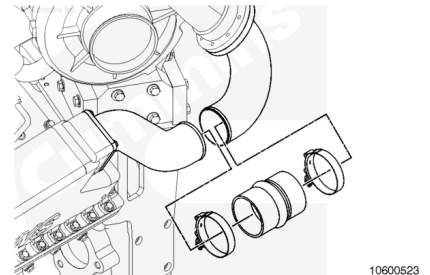
Remove the air crossover support capscrews. Slide the hoses back from the air crossover and remove the air crossover.



Remove the eight capscrews from the air transfer bellows. Remove the bellows and discard the gaskets (marine **only**).

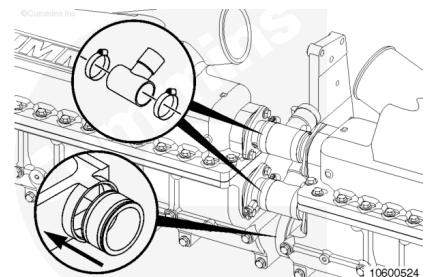


Remove the hose clamps from the air connection tube at the turbocharger connection tube and the aftercooler cover. Slide the hoses onto the connection tube and remove the tube (marine **only**).



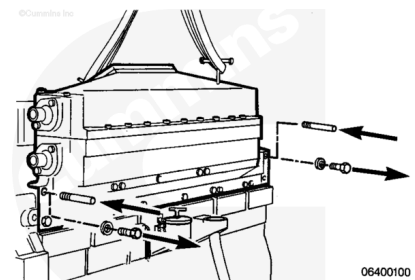
Remove the hose clamps from the rubber tee coolant connections.

Slide the air balance tube toward the adjacent aftercooler housing.



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Remove a lower mounting capscrew at each end of the aftercooler assembly and install guide pins long enough to protrude from the housing.

Remove a cover mounting capscrew at each end of the assembly and install lifting eyes.

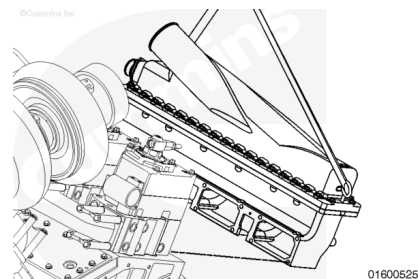
Attach a sling to the lifting eyes and a suitable hoist.

Raise the hoist until tension is applied to the assembly.

Remove the remainder of the mounting capscrews.

Lift and move the assembly away from the cylinder head and the adjacent aftercooler assembly. Be careful not to damage the air balance tube and the rubber coolant connections as the aftercooler assembly is removed.

Repeat the procedure for the remaining assemblies.



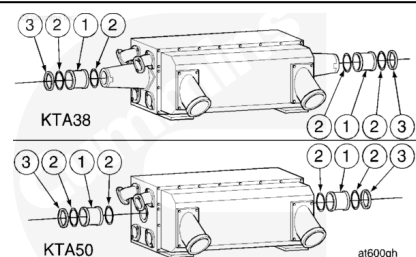
Disassemble

with Mechanically Actuated Injector

Remove the air tubes (1). Remove the o-rings (2). Discard the o-rings.

Remove the dust seals (3). Check the dust seals for excessive oil or hardness. Discard any dust seal that is hard or soaked with oil.

Check for indications of gasket leaks.

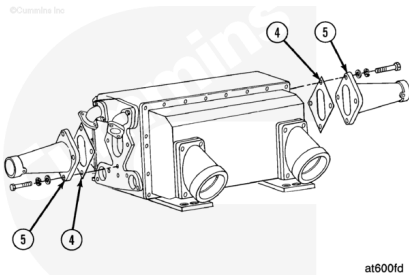


Note : All K50 aftercoolers contain this connection. Newer K38 aftercoolers contain a shorter connection.

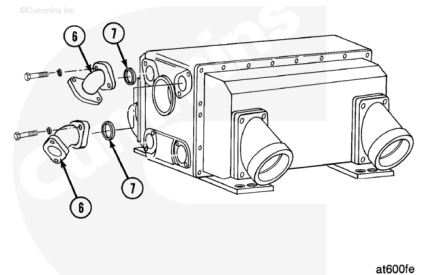
Remove the four capscrews for each crossover connection.

Remove the connections (5) and gaskets (4).

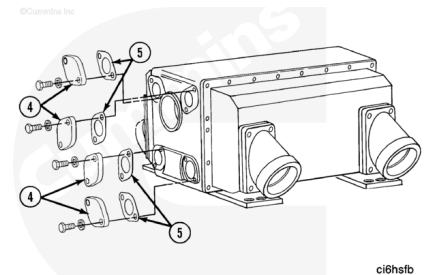
Discard the gaskets.



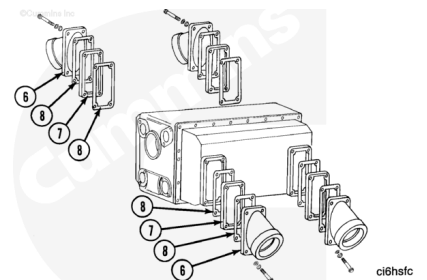
Remove the water outlet connections (6). Remove the rectangular ring or o-ring (7) from the connection. Discard the o-ring.



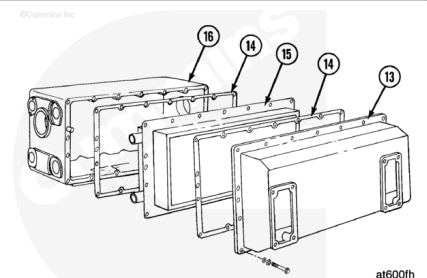
Remove the four cover plates (4) and gaskets (5).
Discard the gaskets.



Remove the four air crossover connections (6), spacers (7), and gaskets (8).
Discard the gaskets.



Remove both of the aftercooler covers (13) and gaskets (14).
Remove the aftercooler elements (15) and the gaskets (14) from the housing (16).



Standard System

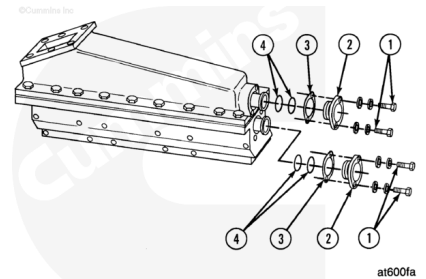
Note : This procedure is performed while the intake manifold (aftercooler housing) remains on the engine.

Remove the parts.

1. Capscrews, lock washers, and flat washers
2. Adapter connections
3. Gaskets.

Discard the gaskets.

Remove the o-rings (6) from the inner adapters (4).

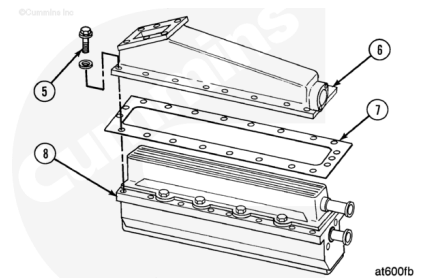


Remove the 21 capscrews and flat washers (5).

Note : Do not damage the gasket surfaces if it is necessary to use a wedge to separate the parts.

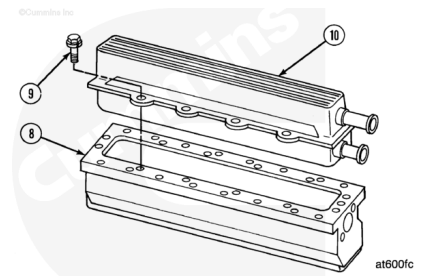
Remove the aftercare cover (6). It will be necessary to move the cover so that it will **not** hit the nipple on the element.

Remove the gasket (7) from the intake manifold (8).



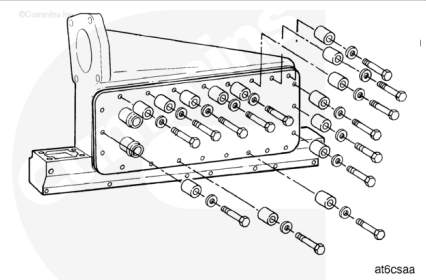
Remove the eight captive capscrews (9). Note that the washers on the capscrews are spring-type washers.

Remove the element (10) from the intake manifold (8).



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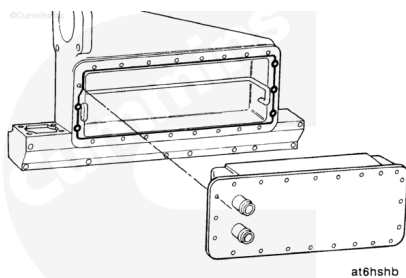


Low Temperature Aftercooler (LTA) System

Note : This procedure is performed while the intake manifold (aftercooler housing) remains on the engine.

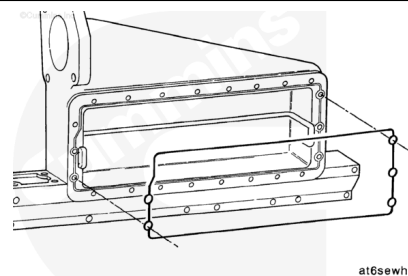
Remove the fourteen capscrews (27), washers (26), and spacers (29).

Remove the aftercooler element from the housing.

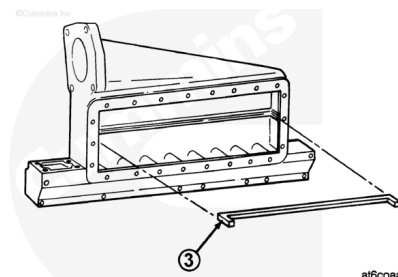


Remove the rubber seal from the groove on the aftercooler housing.

Inspect the seal. Discard if damaged.



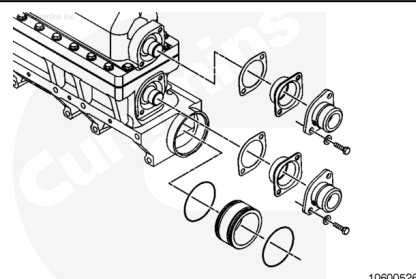
Inspect the rectangular strip seal (3). Discard the seal if it is ripped, cracked, or worn.



with Electronically Actuated Injector

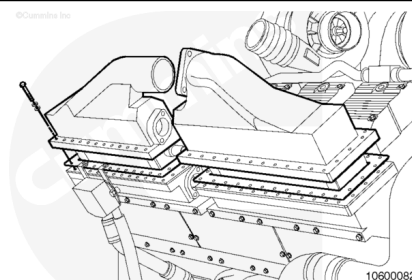
Remove the coolant connections from the aftercooler cover and housing.

Discard the connection gasket.



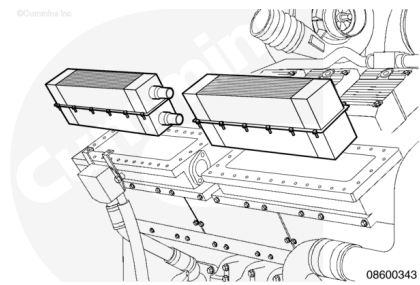
Remove the aftercooler cover mounting capscrews, cover, and gasket.

Discard the gasket, grommet seal, and o-ring seal.



Remove the aftercooler element and gasket.

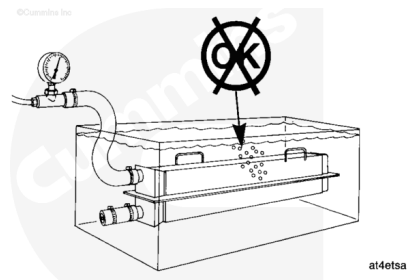
Discard the seals and gaskets.



Pressure Test

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Standard System

- Use a hose plugged with a pipe plug. Plug one of the water nipples as illustrated. Attached an air line and a gauge to the other nipple. Pressure check the element.
- Apply air pressure. Check for leaks.

Measurements

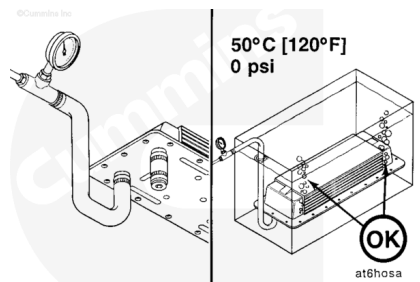
	kpa	psi
Air Pressure	448	65

If the part leaks, it **must** be replaced.

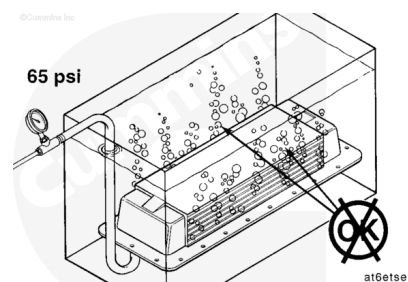
LTA

Low Temperature Aftercooled system

- Use a hose plugged with a pipe plug. Plug one of the element nipples. Attach an air line equipped with a gauge and a regulator valve to the remaining nipple.
- Submerge the core in a tank of water heated to 50° C [120° F]. Do **not** apply air pressure yet. Check the tank for bubbles. Note the location and rate of bubbles. These bubbles are the air entrapped behind the silicone bead.



Leave the element submerged for 5 minutes to allow the temperature of the element to stabilize. Apply air pressure. Check for leaks. The element leaks when bubbles form faster or from a new location while the tank is pressurized.



Measurements

	kpa	psi
Air Pressure	448	65

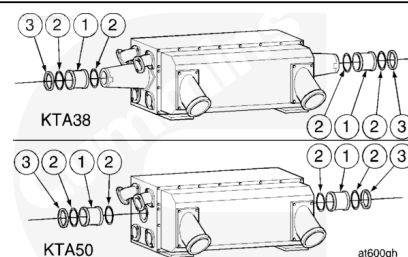
Replace the element if leaks are discovered.

Note : A malfunctioning aftercooler element will often be coated on the outside with coolant deposits.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

All solvent or steam must be drained from the housing. Any liquid remaining in the housing can enter the combustion chamber and damage the engine.

Use solvent or steam.

Clean the exterior of the assembly. Flush the elements through the bores of the water outlet connection at the top. Clean all the bores and grooves for the o-rings. Clean all gasket surfaces. Dry with compressed air.

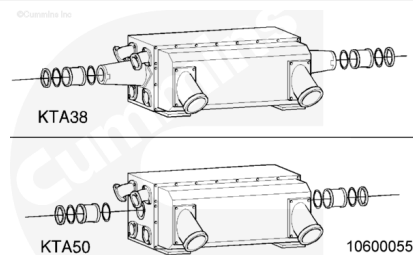
Check the assembly for cracks or other damage. There **must** be no cracks in the housing or cover.

Check the dust seals for excessive oil or hardness. Discard any dust seal that is hard or soaked with oil.

If the parts are cracked or otherwise damaged, or there are any indications of gasket leaks, the assembly **must** be rebuilt.

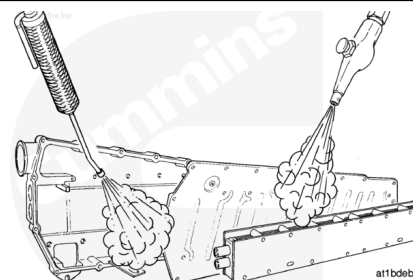
Note : If the K50 aftercooler housing does not contain cross bolts, or contains a rib in the center of the housing, it is recommended that the housing be modified for the cross bolts before it is installed again. Refer to Procedure 010-073 in Section 10.

(/qs3/pubsys2/xml/en/procedures/28/28-010-073.html)



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⚠ WARNING ⚠

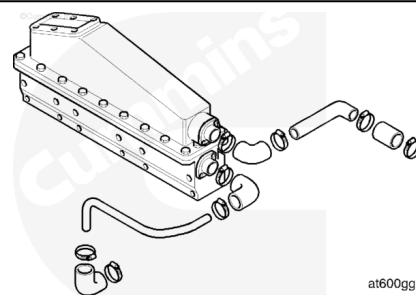
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

Clean all gasket surfaces.

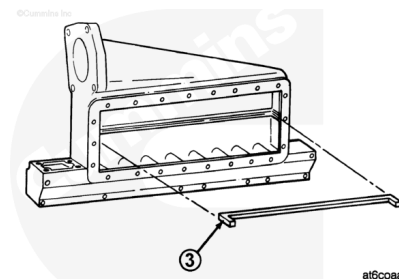
Use solvent or steam to clean the parts.

Check the air passages for evidence of coolant leaks. If there is any evidence of leaks, the assembly **must** be rebuilt.

Check the parts for cracks or other damage.



Inspect the rectangular strip seal (3). Discard the seal if it is ripped, cracked, or worn.



Assemble

with Mechanically Actuated Injector

Center Mount Aftercooler

Install the housing-to-element flange gasket (14). Be sure the proper holes are aligned with the roll pin.

Install the cooler element (15).

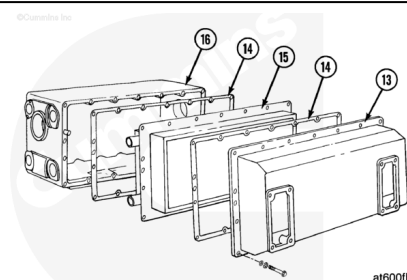
Install a new cover-to-element flange gasket (14) and the cover (13).

Note : Each cover contains 18 capscrews.

Install the flat washers, lock washers, and capscrews.

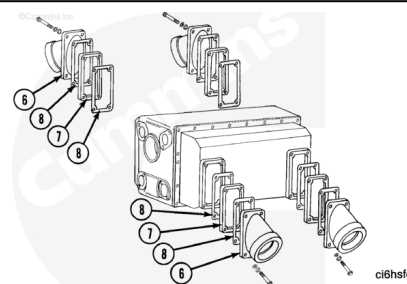
Torque Value: 25 n•m [221 in-lb]

Repeat this step for the other side of the assembly.



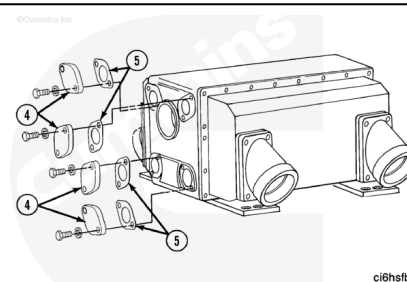
Install the four air crossover connection gaskets (8), the spacers (7), the gaskets (8), and the connection (6).

Torque Value: 25 n•m [221 in-lb]



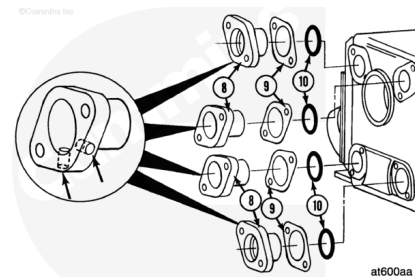
Install the new cover plate gaskets (5) and plates (4).

Torque Value: 25 n•m [221 in-lb]



Note : The adapters must be installed so that the water leakage hole is toward the bottom of the aftercooler housing. This hole prevents any coolant leak from entering the aftercooler housing at the water inlet or outlet connections.

Install four new adapter connection gaskets (9) and adapters (10) on the nipples for the element.



⚠ CAUTION ⚠

Make sure the correct seal is installed for the water connection. If the wrong seal is installed, a leak will result, leading to engine damage.

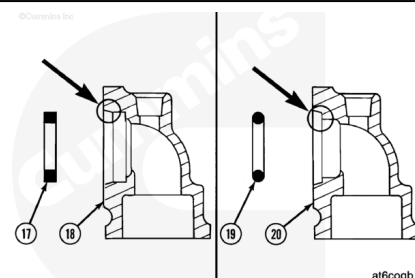
Note : This information applies to the water inlet and outlet connections for the aftercooler.

If necessary, install a new rectangular seal ring or a new o-ring on the water outlet connection. Look at the water outlet connection.

Install the rectangular ring seal (17) with the connection (18) that does **not** have the step at the end of the chamfer.

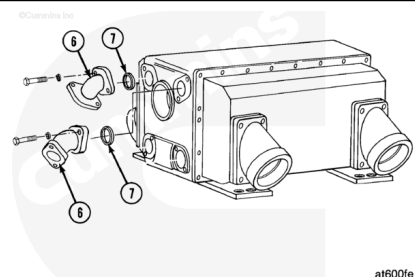
Install the o-ring (19) with the connection (20) that has a step at the end of the chamfer.

Note : The rectangular seal ring is the same part number as the seal on the water outlet tube. The o-ring is the same part number as the seal (10) on the adapter connection (8). The older connection (20) is **not** available for service.



Install both water outlet connection seals (7) and connections (6). Install the lock washers and capscrews.

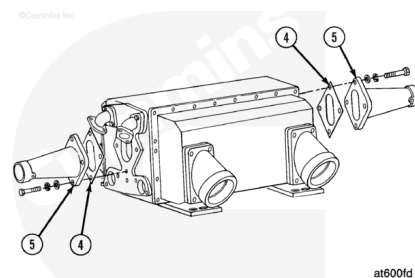
Torque Value: 25 n•m [221 in-lb]



Note : All K50 aftercoolers contain this connection. Newer K38 aftercoolers contain a shorter connection.

Install new air crossover connection gaskets (4) and connections (5). Install the flat washers, lock washers, and capscrews.

Torque Value: 25 n•m [221 in-lb]



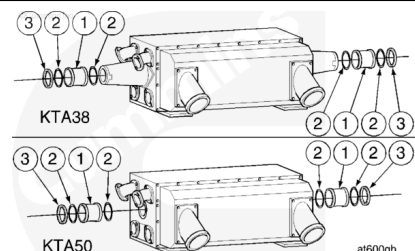
Note : Lubricate **only** the o-ring seals that are to be installed immediately in the bore of the housing or connection. This will prevent the exposed o-ring from expanding and collecting dirt.

Install two new o-rings (2) on each air tube.

Install the dust seals (3) on the air tubes (1).

Use vegetable oil. Lubricate the o-ring.

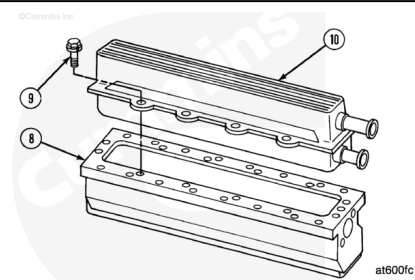
Install the air tube in the bore of the housing or connection.



Standard System

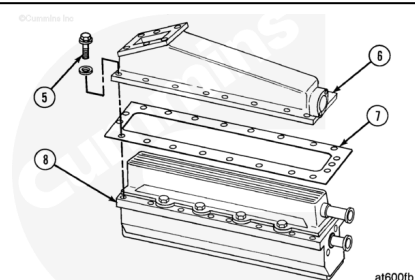
Install the element (10) in the manifold (8). Install the eight capscrews (9) that contain the captive spring washers.

Torque Value: 25 n•m [221 in-lb]



Install a new gasket (7) and cover (6). Install the 21 captive capscrews and flat washers (5).

Torque Value: 25 n•m [221 in-lb]



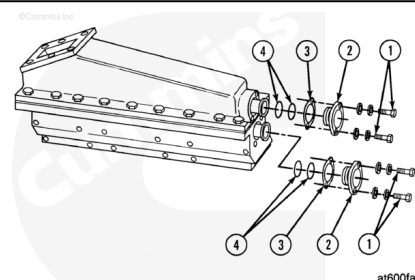
Install two new o-rings (4) in each adapter (2).

Use vegetable oil. Lubricate the o-rings.

Install new gaskets (3) and the adapters (2).

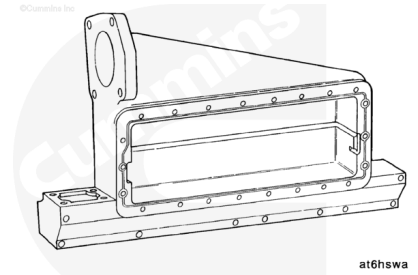
Install the four capscrews with flat washers and lock washers (1).

Torque Value: 16 n•m [142 in-lb]



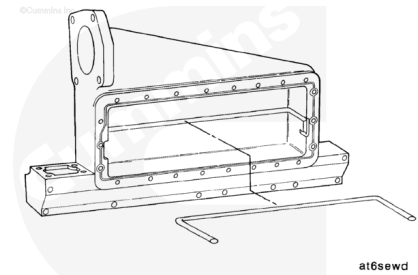
Low Temperature Aftercooler (LTA)

The intake manifolds serve as the aftercooler housing. The element is installed through the cast opening in the side of the housing.



⚠ WARNING ⚠

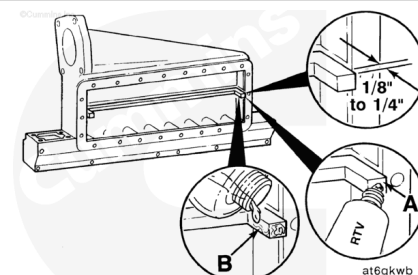
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



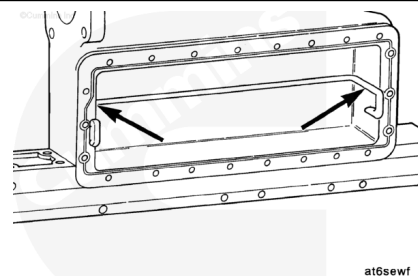
Install the foam seal into the groove in the aftercooler housing.

Note : Applying a small amount of sealant (RTV) into the groove aids in holding the foam seal in place. Do **not** allow sealant on the visible portion of the foam seal.

The length of the seal protruding from each end is then bent and installed into the groove provided to obtain a positive seal with the element mounting flange. The protrusion **must** be 3 to 6 mm [0.118 to 0.236 inch] outside the housing at both sides. If required, cut the length to this specification.



Apply a light film of vegetable oil to the foam seal at the ends of the housing to reduce drag of the element and prevent damage to the foam seal during element installation.



There are two methods of sealing the aftercooler to element joint:

- Gasket/Sealant
- Rubber Seal.

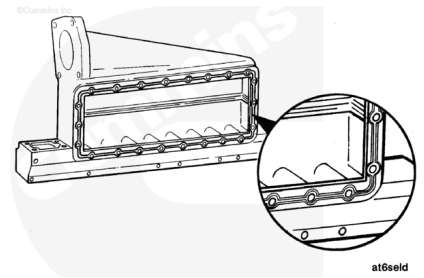
Gasket/Sealant Installation

Note : The gasket was replaced with liquid sealant.

Apply two separate, continuous beads of liquid sealant Loctite® 509 (also called Loctite® LMD127) directly to the aftercooler housing. Each sealant bead is to be 1.5 mm [0.039 inch] in diameter. This sealant replaced the conventional gasket.

The sealant is an anaerobic substance and will **only** cure (set up) in the absence of air. The sealant will **not** begin to cure until the aftercooler element is clamped into the housing.

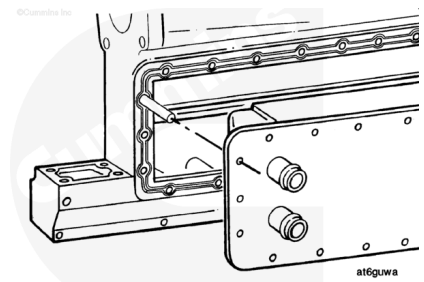
The sealant beads **must** be continuous strips without gaps, to prevent air leakage. Do **not** apply an excessive amount of sealant. Extra sealant does **not** result in a stronger joint.



Install two guide pins in the housing for installing the aftercooler element into the housing. The guide plugs allow installation of the heavy element without disturbing the sealant, if used.

Guide pins can be made by cutting the head off a 3/8 inch capscrew.

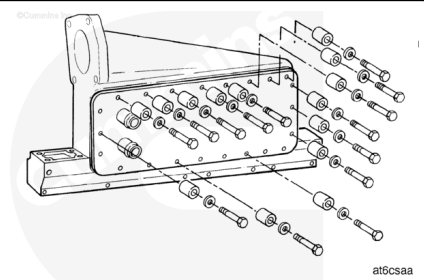
Install the aftercooler element into the aftercooler housing.



Install the 63 mm [2.48 inch] capscrews and washers into the locations shown.

Tighten the capscrews just enough to hold the element in place.

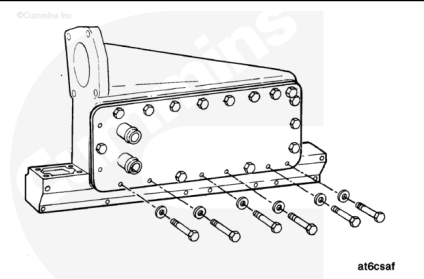
Remove the two guide pins.



Install the 171 mm [6.732 inch] capscrews and washers into the locations as shown.

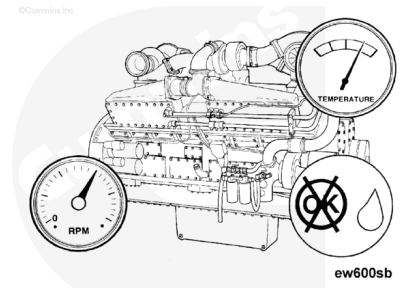
Tighten the capscrews just enough to hold the element in place.

The two remaining holes will be used when the water transfer connections are installed. Use the torque value and sequence in this step.



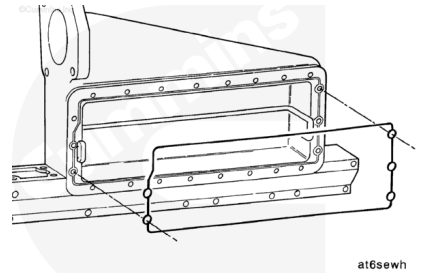
The sealant cures to 80 percent strength 1 hour after the aftercooler element capscrews are tightened. The sealant is completely cured in 24 hours.

Once the engine is filled with coolant and the engine is tested, a small amount of sealant will seep from the joint during the first hour of engine operation. Remove this excess sealant.



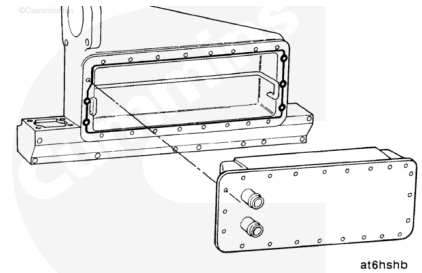
Rubber Seal Installation

Install the rubber seal into the groove on the aftercooler housing. Make sure the seal is **not** twisted or damaged.



Install the aftercooler element into the housing. Make sure **not** to dislocate either the foam seal or the rubber seal.

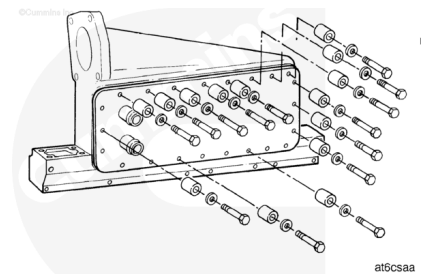
All aftercooler elements are identical and are inserted into the housing with the water inlet and outlet just below the air intake connection.



Install the washers (26) and spacers (29) on the fourteen capscrews (27). Insert the capscrews into the holes shown. The three capscrews (1) can only be installed in the rear intake manifold assemblies at this time.

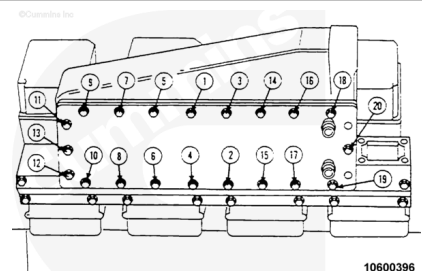
Capscrews will be inserted in the open positions either during the installation of the aftercooler plumbing or the intake manifold assembly.

Tighten all of the inserted capscrews just enough to hold the aftercooler element in place.



Use the sequence shown. Tighten the capscrews.

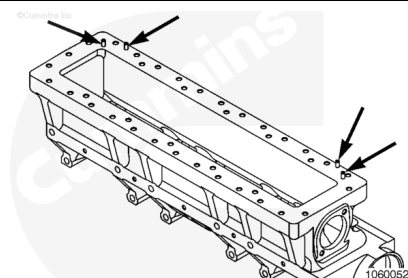
Torque Value: 45 n•m [33 ft-lb]



with Electronically Actuated Injector

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

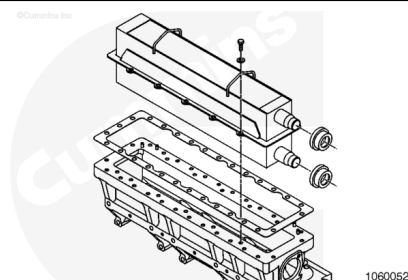


Check the four aftercooler element locating pins for proper location.

Install the aftercooler element gasket.

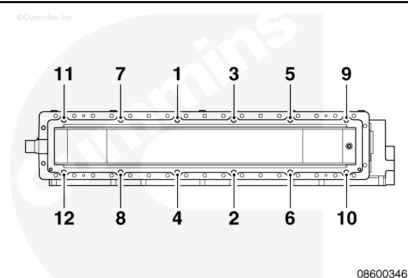
Install the upper and lower grommet seals and o-ring seals on the cooler connections.

Install the aftercooler element into the housing.



Install the twelve capscrews and tighten in the sequence illustrated.

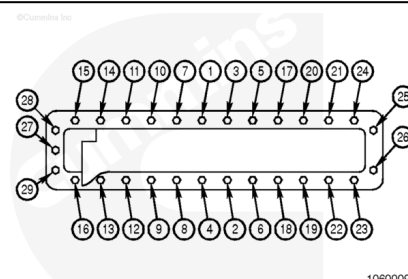
Torque Value: 45 n•m [33 ft-lb]



Apply a light coat of anti-seize lubricant, Part Number 3824759, to the aftercooler cover mounting capscrew threads.

Install the aftercooler cover and mounting capscrews. Tighten the capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]

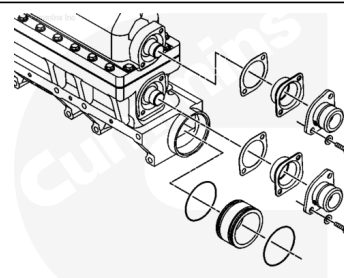


Lubricate the o-ring with Lubriplate™ 105, Part Number 3163087, multi-purpose lubricant

Install the connection gasket and the seal retainer.

Install the flange connector and the mounting capscrews.

Tighten the capscrews.



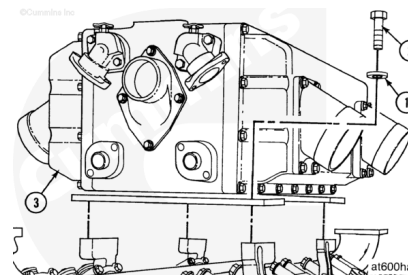
Torque Value: 30 n•m [22 ft-lb]

Install

with Mechanically Actuated Injector

⚠ WARNING ⚠

This assembly or component weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : This procedure applies only to KTA38 and KTA50 engines with a center mount aftercooler. If a KT38 engine is being serviced, omit this procedure and proceed to install the air transfer housing. If an engine with outboard aftercoolers is being serviced, proceed to Install the Turbochargers.

Note : This procedure assumes that the aftercooler was assembled previously. Refer to Procedure 010-999 in Section F. (</qs3/pubsys2/xml/en/procedures/28/28-010-999-tr.html>)

Note : Newer engines have three capscrews with lock washers through the bracket to each support. Older engines have two capscrews with lock washers.

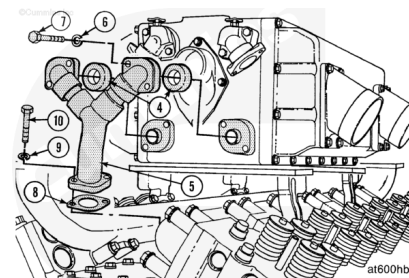
Center Mount Aftercooler

Install the lock washers (1) and capscrews (2) in the holes in the bracket. Put the aftercooler assembly (3) on the supports and install the capscrews in the supports.

Torque Value: 60 n•m [44 ft-lb]

⚠ CAUTION ⚠

Newer engines have rectangular ring seals (4) between the water inlet connections and the tubes on the aftercooler elements. Older engines have o-rings. Determine if seals are required. If the correct seals are not installed, a leak could result leading to engine damage.



Note : Assemble the aftercooler water inlet connection (5) as shown. Refer to Procedure 010-999 in Section F. (/qs3/pubsys2/xml/en/procedures/28/28-010-999-tr.html)

Install the seals (4) on the aftercooler element tubes.

Use vegetable oil. Lubricate the seals.

Install the water inlet connection (5). Push the connection over the seals until the connection touches the aftercooler housing. Install, but do **not** tighten, the four lock washers (6) and capscrews (7).

Install the gasket (8), the lock washers (9), and the capscrews (10).

Tighten the four capscrews (7).

Torque Value: 25 n•m [221 in-lb]

Tighten the two capscrews (10).

Torque Value: 40 n•m [30 ft-lb]

If the turbocharger uses a hose connection, slide the hose forward on the connection and tighten the **T-bolt** clamps.

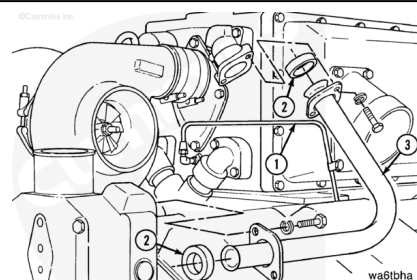
Torque Value: 9 n•m [80 in-lb]

Note : Installation of the air fuel control air tube (1) is not as difficult if completed before the installation of the water tubes.

Note : Two aftercooler water outlet tubes are required

Install two rectangular seal rings (2) on the water outlet tube. Install the tube, lock washers, and capscrews.

Torque Value: 40 n•m [30 ft-lb]

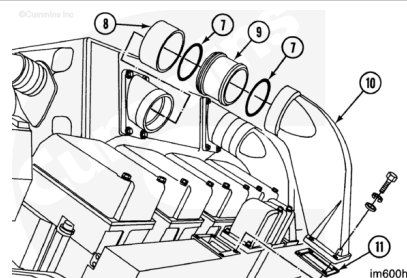


Note : Engines that are equipped with a center mount aftercooler have four air crossover assemblies. Use this procedure to install the parts.

Install the two o-rings (7) and the dust seal (8) on the tube (9) for the air crossover.

Use engine oil or Lubriplate™ 105, Part Number 3163087, multi-purpose lubricant. Lubricate the seals.

Install the tube in the air connections on the aftercooler or the air transfer housing.



Install the air crossover (10) on the tube. Push the crossover on the tube as far as possible.

Note : The raised bead on the gasket must be toward the air crossover.

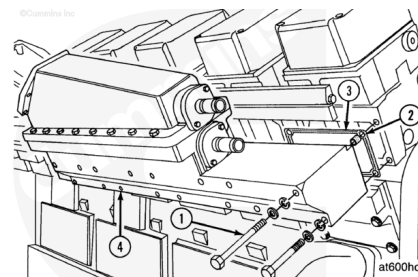
Install the gasket (11) and the four lock washers, flat washers, and capscrews.

Torque Value: 25 n•m [221 in-lb]

Outboard Aftercooler - Standard Cooling Systems

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the long capscrews, lock washers, flat washers (1) and the rectangular seal sleeves (2) in the row of capscrew holes that is nearest to the top of the aftercooler assembly. The sleeves prevent dirt and water from damaging the threads of the capscrews.

Note : The use of guide pins installed in the row of capscrew holes at the bottom of the cylinder head will aid the assembly procedure.

Note : The raised bead of the gasket (3) must be toward the cylinder head.

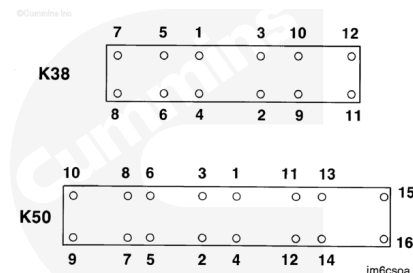
Install a gasket (3) on each cylinder head.

Install the aftercooler assembly (4), lock washers, flat washers, and the capscrews.

Use the following steps to tighten the capscrews in the sequence shown.

Torque Value:

1. 25 n•m [221 in-lb]
2. 45 n•m [33 ft-lb]



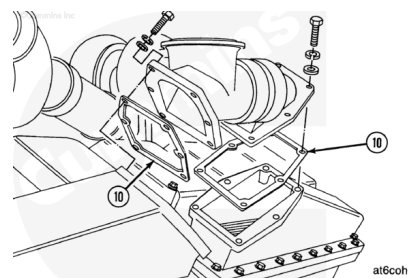
Type 1- Air Crossovers with Retaining Straps

Note : The engine has two air jumper tube assemblies. The assembly for the left bank is shown. The assembly

for the right bank **must** be installed so that the opening for the air crossover is nearest to the rear of the engine.

Install the two gaskets (10) and the air jumper tube assembly with the 12 flat washers, 12 lock washers and 12 capscrews.

Torque Value: 25 n•m [221 in-lb]



Install two o-ring seals (11) on the tube (12) for the air crossover. Install the dust seal (13) on the tube.

Use engine oil or Lubriplate™ 105, Part Number 3163087, multi-purpose lubricant. Lubricate the seals.

Install the tube in the turbocharger. Install the air crossover (14) on the tube. Push the crossover on the tube as far as possible.

Note : Check to be sure the crossover and turbocharger compressor housing are aligned properly. If the gaps between the compressor housing and the crossover are **not** the same at the top and the bottom, adjust the turbocharger compressor housing alignment.

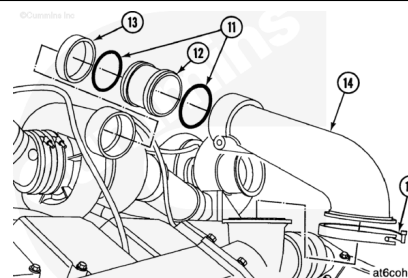
Install the V-band clamp (15). Tighten the clamp.

Torque Value: 9 n•m [80 in-lb]

Use a mallet. Tap around the clamp to be sure it is seated. Tighten the clamp nut.

Torque Value: 9 n•m [80 in-lb]

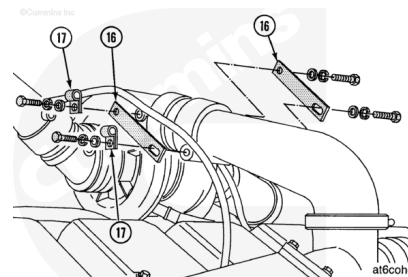
Continue tapping and tightening the clamp nut until it does **not** turn at the specified torque after tapping.



Note : Some engines have clamps (17) for the turbocharger oil inlet hose. The hose **must** be clamped to the outside of the retainer plates (16).

Install the two retainer plates (16), the large flat washers, lock washers, and capscrews.

Torque Value: 6 n•m [53 in-lb]



Type II - Air Crossover with Hoses

Install the hose (12) on the connection (13). Push the hose on the connection as far as possible.

Insert the connection and the hose between the two manifold assemblies. Fit the hose over the manifold cover.

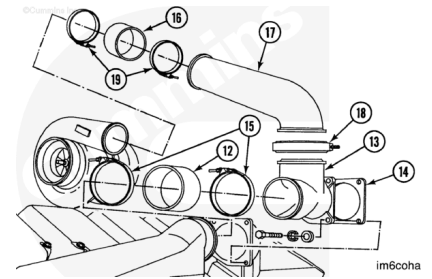
Install the gasket (14), the four flat washers, lock washers, and capscrews.

Torque Value: 25 n•m [20 in-lb]

Push the hose on the manifold cover until the ends of the hose are the same distance from the gap between the connection and the cover.

Install the two T-bolt clamps (15).

Torque Value: 9 n•m [80 in-lb]



Note : Check to be sure the crossover and the turbocharger compressor housing are aligned properly. Put the crossover in position on the connection (13). Look at the gap between the crossover and the turbocharger housing. If the gap is **not** the same at the top and bottom, the turbocharger compressor housing **must** be adjusted

Install the hose (16) on the crossover (17). Install the hose and tube on the turbocharger housing.

Install the V-band clamp (18).

Torque Value: 9 n•m [80 in-lb]

Use a mallet. Tap around the clamp to be sure it is seated.

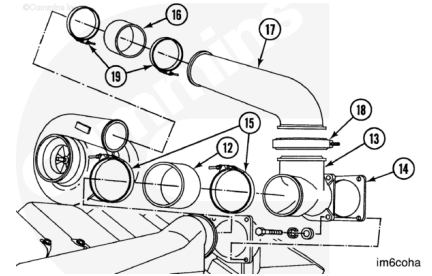
Tighten the nut on the clamp.

Torque Value: 9 n•m [80 in-lb]

Continue tapping and tightening the clamp nut until it does **not** turn at the specified torque after tapping.

Install the two T-bolt clamps (19).

Torque Value: 8 n•m [71 in-lb]



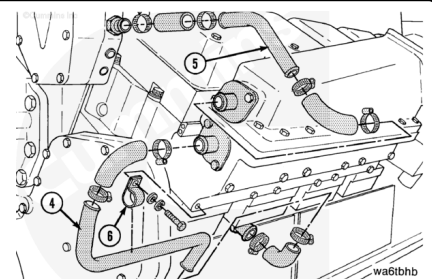
Old Style Plumbing with Hoses

Install the hoses and clamps on the inlet tube 4 and the outlet tube (5). Connect the tubes to the aftercooler, the fitting in the cylinder block and the fitting in the thermostat support as shown.

Install the support clamp (6) on the tube with one of the aftercooler mounting capscrews.

Note : Only one of the four connections is shown here.

Torque Value: 5 n•m [44 in-lb]



New Style Plumbing

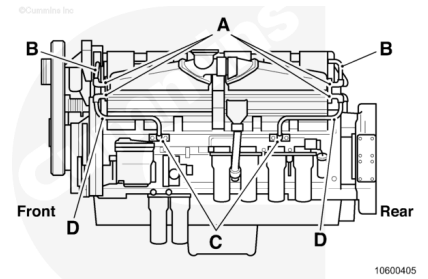
Install the hoses on the water transfer tube (C). Install the o-rings and rectangular seal in the core connections.

Lubricate with engine oil or vegetable oil.

Tighten the connecting capscrews.

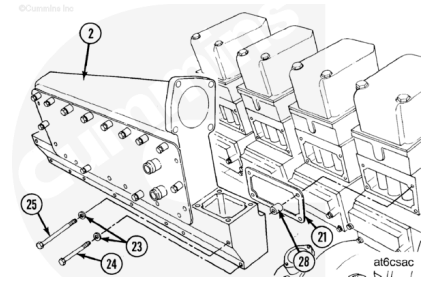
Torque Value: 25 n·m [221 in-lb]

Install the clamps. Install inlet tubes (B).



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Outboard Aftercooler - LTA Cooling System

Note : The aftercooler housings for the left bank front and right bank rear are the same. The left bank rear and the right bank front are also the same.

Note : Guide pins installed in the row of capscrew holes at the bottom of the cylinder heads will aid in installation.

Note : The raised bead of the gasket (21) must be toward the cylinder head.

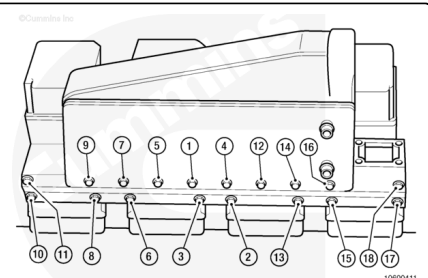
Install the aftercooler assembly (2), conical washers (23) and the capscrews (25).

Install the longer capscrews (25), conical washers (23), and the rectangular seal sleeves (28) in the row of capscrews holes that are nearest the top of the aftercooler assembly. The sleeves will prevent dirt and water from damaging the threads of the capscrews.

Insert the shorter capscrews (24) and conical washers (23) in the lower row of the assembly.

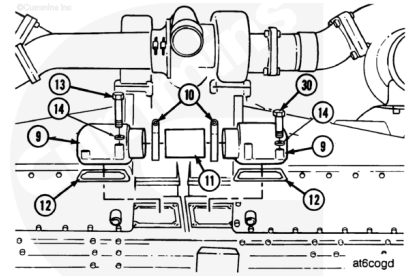
Tighten the capscrews in the sequence shown.

Torque Value: 45 n·m [33 ft-lb]



Install the hose (11) and the two T-bolt clamps (10) on the two air intake connections (9).

Install gaskets (12) and connections on the intake manifolds. Insert the long capscrews (13) and the short capscrews (30) with washers (14). Tighten the capscrews (13) and (30).



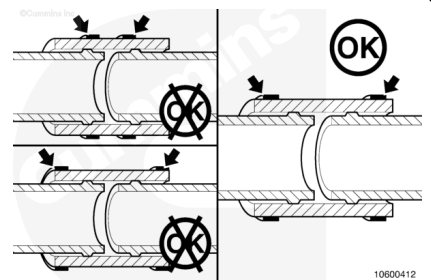
Torque Value: 25 n•m [221 in-lb]

Tighten the T-bolt clamps.

Torque Value: 8 n•m [70 in-lb]

Do **not** position the clamps over the hose beads. The clamps **must** be positioned between the hose beads and the end of the hose. A portion of the hose **must** extend outside the clamp.

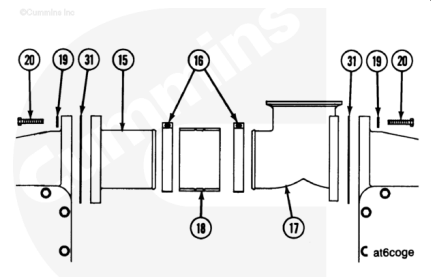
Tighten the T-bolt clamps.



Torque Value: 8 n•m [71 in-lb]

Install the two clamps (16) and the hose (18) over the connections (15 and 17). Do **not** tighten the clamps at this time.

Install the gaskets (31), eight capscrews (20) and washers (19). Tighten the capscrews.

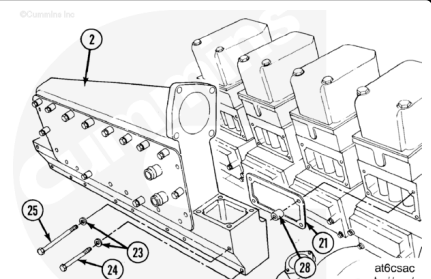


Torque Value: 40 n•m [30 ft-lb]

Center the hose (18) over the gap between the connections (15 and 17).

Note : Do **not** position the clamps over the hose beads. The clamps **must** be positioned between the hose beads and the end of the hose. A portion of the hose **must** extend outside the clamp.

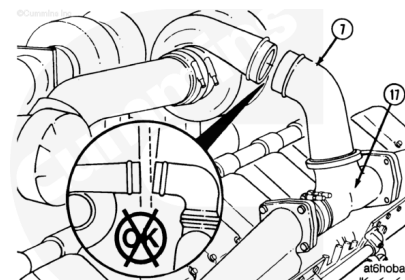
Position the clamps between the end of the hose and bead on the connection. Tighten the clamps.



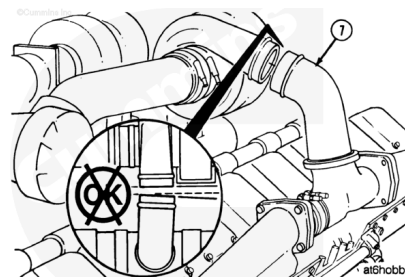
Torque Value: 8 n•m [71 in-lb]

The crossover (7) and the turbocharger **must** align properly. Put the crossover in position on the connection (17). Check the gap between the crossover (7) and the turbocharger housing. If the gap is **not** the same at the top and bottom,

rotate the crossover connection (17). If this does **not** provide proper alignment, the turbocharger compressor housing **must** be adjusted.



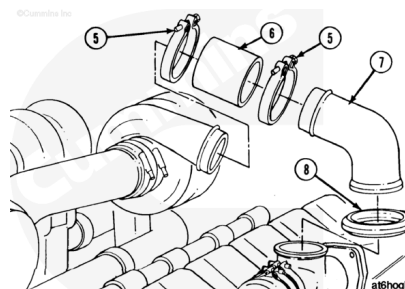
If the gap is **not** the same at the front and rear, rotate the crossover to the position that provides the best possible alignment.



Install the hose (6) on the crossover (7). Then install the crossover on the turbocharger and the flanged connection.

Install and tighten the V-band clamp (8).

Torque Value: 9 n•m [80 in-lb]

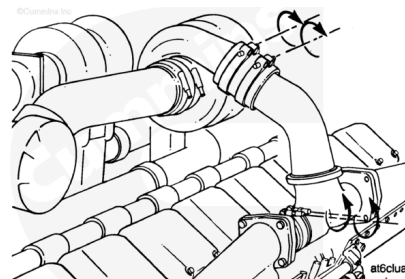


Tighten the nut on the clamp.

Torque Value: 9 n•m [80 in-lb]

Tap and tighten the clamp nut until it will **not** turn at the specified torque after tightening.

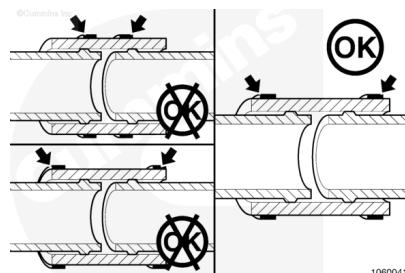
Install the two T-bolt clamps.



Note : Do not position the clamps over the hose beads. The clamps **must** be positioned between the hose beads and the end of the hose. A portion of the hose **must** extend outside the clamp

Position the hose (6) in the center of the connection and tighten the clamps (5).

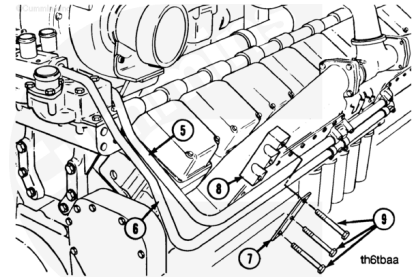
Torque Value: 8 n•m [71 in-lb]



Install the tube clamps (7) and (8) on water transfer tubes (5) and (6). Install capscrews (9).

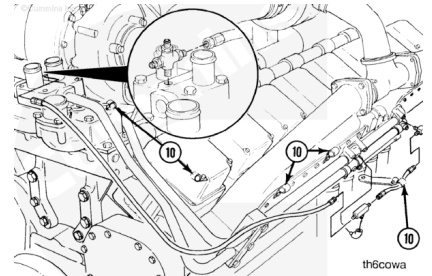
Torque Value: 45 n·m [33 ft·lb]

Check dust seals (29).

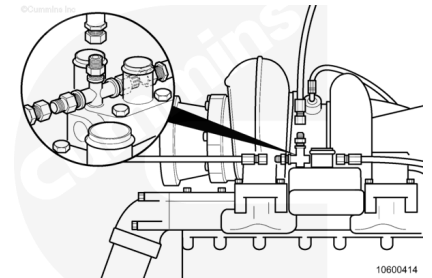


Install the vent line on the upper water transfer connection on each bank.

Note : The lines must route horizontally or inclined toward the thermostat housing at all times. Any humps or sags will prevent aftercooler core venting and cause core damage. The length of the short hose between the two manifolds is critical. It must not sag or put excessive stress on the fittings



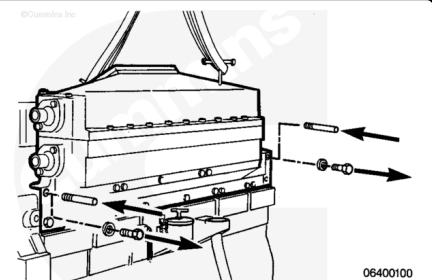
Install the vent lines from each bank to the cross connection on the thermostat housing.



with Electronically Actuated Injector

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use a hoist and sling to lift the aftercooler assembly.

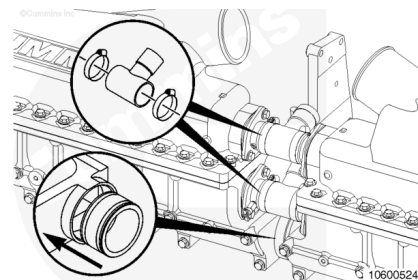
Install guide studs long enough to protrude from the housing.

Install guide studs into the cylinder head in the lower row of mounting holes.

Lubricate the air balance tube o-rings with Lubriplate™ 105, Part Number 3163087, multi-purpose lubricant

Install the air balance tube.

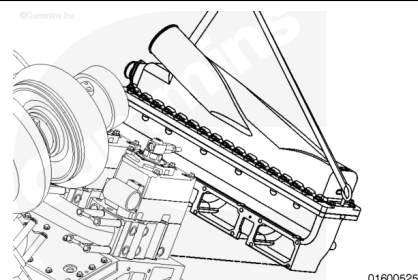
Install the rubber T-coolant connectors.



Install new intake manifold gaskets, with the raised bead of the gasket toward the cylinder head. Use heavy grease to hold in place.

Install the aftercooler assembly with a hoist.

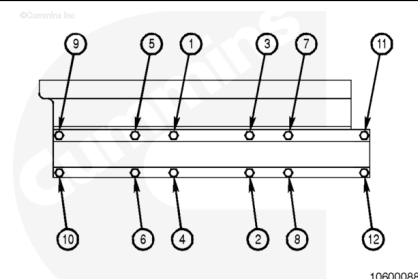
Align the rubber water connectors and air balance tube while the assembly is being installed over the guide studs.



Install the top row of the aftercooler mounting capscrews (long capscrews). Do **not** tighten the capscrews.

Remove the guide studs and install the lower aftercooler mounting capscrews.

Tighten the aftercooler mounting capscrews in the sequence shown.



Torque Value:

Cast Iron Housing Capscrews
1. 65 n•m [48 ft-lb]

Torque Value:

Aluminum Housing Capscrews
1. 53 n•m [39 ft-lb]

Install and tighten the coolant connection hose clamps.

Torque Value:

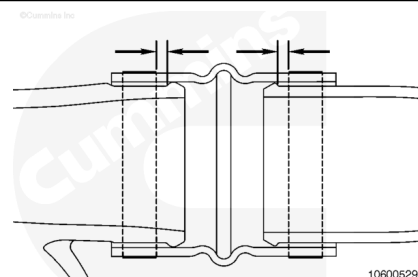
Coolant Connection Clamps
1. 5 n•m [44 in-lb]

Inspect the air crossover, turbocharger connection, and the hoses to make sure they are clean.

Install the air crossover, hoses, and the hose clamps.

Install the air crossover support capscrews.

Position the hose clamps 10 mm [0.25 in] from the raised bead on the turbocharger outlet and the air crossover.



Torque Value: 9 n•m [80 in-lb]

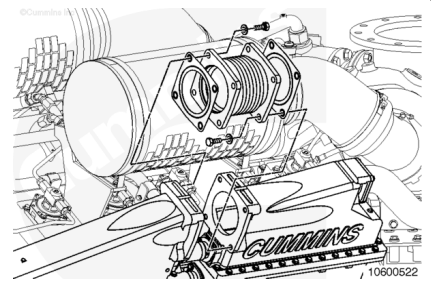
Tighten the air crossover support capscrews.

Torque Value: 45 n•m [33 ft-lb]

Install the air connection bellows and two new gaskets (marine **only**).

Install and tighten the eight mounting capscrews.

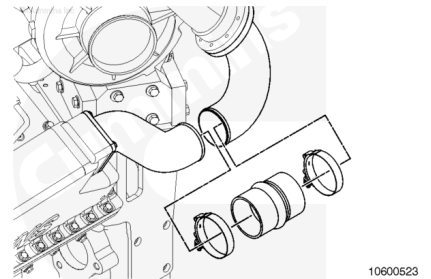
Torque Value: 142 n•m [105 ft-lb]



Install the air transfer tube between the aftercooler and the turbocharger (marine **only**).

Install and tighten the hose clamps.

Torque Value: 11 n•m [97 in-lb]



Finishing Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the batteries. See equipment manufacturer service information.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-

018-tr.html)

- Operate the engine and check for leaks.

with Electronically Actuated Injector

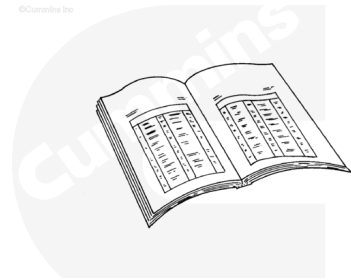
⚠ WARNING ⚠

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⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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- Install the exhaust gas temperature converters, if equipped. Refer to Procedure 019-450 in Section 19 in the Troubleshooting and Repair Manual QSK38, QSK 50, and QSK60 Modular Common Rail System, Bulletin 4021533. (</qs3/pubsys2/xml/en/procedures/05/05-019-450.html>)
- Install the intake manifold air temperature sensor. Refer to Procedure 019-059 in Section 19 in the Troubleshooting and Repair Manual QSK38, QSK 50, and QSK60 Modular Common Rail System, Bulletin 4021533. (</qs3/pubsys2/xml/en/procedures/05/05-019-059.html>)
- Install the intake manifold air pressure sensor. Refer to Procedure 019-061 in Section 19 in the Troubleshooting and Repair Manual QSK38, QSK 50, and QSK60 Modular Common Rail System, Bulletin 4021533. (</qs3/pubsys2/xml/en/procedures/05/05-019-061.html>)
- Install the coolant bypass tube. Refer to Procedure 008-005 in Section 8 (</qs3/pubsys2/xml/en/procedures/28/28-008-005-tr.html>) (right bank **only**).
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks.

Last Modified: 03-Mar-2016
